Merged Solutions: Exercises 1.1–1.3, 1.5–1.7 and 2.1–2.8, 3.1

July 29, 2026

Exercise 1.1

(a) Show that the inner product has the following properties for all $x, y, z \in \mathbb{R}^n$ and $a \in \mathbb{R}$:

$$\langle x, y\rangle = \langle y, x\rangle, \qquad \langle x+y, z\rangle = \langle x, z\rangle + \langle y, z\rangle, \qquad \langle ax, y\rangle = a\langle x, y\rangle.$$

Solution. These follow from the Euclidean inner product $\langle x, y\rangle = \sum_{i=1}^n x_i y_i$.

(b) For $t \in \mathbb{R}$ and $x, y \in \mathbb{R}^n$ show

$$\|x+ty\|^2 = \|x\|^2 + 2t\langle x, y\rangle + t^2\|y\|^2 \geq 0.$$

Solution. Expand $\langle x+ty, x+ty\rangle$.

(c) Deduce Cauchy–Schwarz: $|\langle x, y\rangle| \leq \|x\|\,\|y\|$.
Solution. View $\|x+ty\|^2$ as a quadratic in t and use non-positivity of its discriminant. Equality iff x, y are linearly dependent.

(d) Deduce the triangle inequality $\|x+y\| \leq \|x\| + \|y\|$.

(e) Show the reverse triangle inequality $|\|x\| - \|y\|| \leq \|x-y\|$.

(f) Suppose $x = (x^1, \ldots, x^n)^t$.

(i) $\max\limits_k |x^k| \leq \|x\|$.

(ii) $\|x\| \leq \sqrt{n}\,\max\limits_k |x^k|$.

Solution. Immediate from $\|x\|^2 = \sum (x^i)^2$.

Exercise 1.2

Let (x_i) and (y_i) in $\mathbb{R}^n$ satisfy $x_i \to x$, $y_i \to y$.

(a) $x_i + y_i \to x + y$. *Proof.* $\|(x_i+y_i) - (x+y)\| \leq \|x_i - x\| + \|y_i - y\| \to 0$.

(b) $\langle x_i, y_i\rangle \to \langle x, y\rangle$; in particular $\|x_i\| \to \|x\|$.
Proof. Expand $\langle x_i, y_i\rangle - \langle x, y\rangle$ and use Cauchy–Schwarz.

(c) If $a_i \to a$ in $\mathbb{R}$, then $a_i x_i \to ax$.
Proof. $a_i x_i - ax = (a_i - a)(x_i - x) + (a_i - a)x + a(x_i - x)$.

Exercise 1.3

Which of the following subsets of $\mathbb{R}^n$ is open?

(a) $\mathbb{R}^n$: **Yes**.

(b) $\varnothing$: **Yes**.

(c) $\{x \in \mathbb{R}^n : x^1 > 0\}$: **Yes**.

(d) $\{x \in \mathbb{R}^n : x^i \in [0,1]\}$: **No**.

(e) $\mathbb{Q}^n$: **No**.

Exercise 1.5

(a) If $x_i \to x$ and $\|x_i\| < r$ for all i, then $\|x\| \leq r$.
Proof. Contradict $\|x\| > r$ using $\varepsilon = (\|x\| - r)/2$ and the reverse triangle inequality.

(b) The closure of $B_r(y) = \{x : \|x - y\| < r\}$ is $\overline{B_r(y)} = \{x : \|x - y\| \leq r\}$.
Proof. Use (a) for the first inclusion; for boundary points take $x_i = y + (1 - \frac{1}{i})(x - y)$.

Exercise 1.6

Find A°, $\overline{A}$ and ∂A.

(a) $A = \mathbb{R}^n$: $A^\circ = \mathbb{R}^n$, $\overline{A} = \mathbb{R}^n$, $\partial A = \varnothing$.

(b) $A = \{0\}$: $A^\circ = \varnothing$, $\overline{A} = \{0\}$, $\partial A = \{0\}$.

(c) $A = \{x : x^1 \geq 0\}$: $A^\circ = \{x : x^1 > 0\}$, $\overline{A} = A$, $\partial A = \{x : x^1 = 0\}$.

(d) $A = \{x : x^i \in [0,1]\ \forall i\}$: $A^\circ = \{x : x^i \in (0,1)\ \forall i\}$, $\overline{A} = A$, $\partial A = A \setminus A^\circ$.

(e) $A = \{x : x^i \in \mathbb{Q}\ \forall i\}$: $A^\circ = \varnothing$, $\overline{A} = \mathbb{R}^n$, $\partial A = \mathbb{R}^n$.

Exercise 1.7

(a) If U_1, U_2 are open, then $U_1 \cup U_2$ and $U_1 \cap U_2$ are open.

(b) If E_1, E_2 are closed, then $E_1 \cup E_2$ and $E_1 \cap E_2$ are closed.

(c) For any collection $\mathcal{U}$ of open sets: $\bigcup \mathcal{U}$ is open; $\bigcap \mathcal{U}$ need not be open (e.g. $\cap_{n \geq 1}(-1/n, 1/n) = \{0\}$). For closed sets: arbitrary intersections are closed, finite unions are closed, infinite unions need not be closed.

Exercise 2.1

A sequence $(x_i) \subset \mathbb{R}^n$ is Cauchy if $\forall \varepsilon > 0\, \exists N : \ i, j \geq N \Rightarrow \|x_i - x_j\| < \varepsilon$.

(a) If $x_i \to x$, then (x_i) is Cauchy. *Proof.* Triangle inequality with $\varepsilon/2$.

(b) If (x_i) is Cauchy, then it is bounded. *Proof.* Take $\varepsilon = 1$.

(c) Every Cauchy sequence in $\mathbb{R}^n$ converges. *Proof.* Bounded $\Rightarrow$ has a convergent subsequence; use Cauchy property to pass to the full sequence.

Remark. In $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{C} \cong \mathbb{R}^2$, a sequence is convergent iff it is Cauchy (completeness). In incomplete spaces (e.g. $\mathbb{Q}$), Cauchy does not imply convergent.

Exercise 2.2

Suppose $A \subset \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Then

$$\lim_{x \to p} f(x) = F \iff \text{for every sequence } x_i \in A,\ x_i \neq p,\ x_i \to p,\ \text{we have } f(x_i) \to F.$$

Proof. The forward direction is immediate from the ε–δ definition; the reverse is proved by contradiction constructing a sequence $x_k \to p$ with $f(x_k)$ staying ε_0 away from F.

Exercise 2.3

(a) $f : \mathbb{R} \to \mathbb{R}^n$, $x \mapsto (x, 0, \ldots, 0)^t$ is continuous: $\|f(x) - f(x_0)\| = |x - x_0|$.

(b) For $A \subset \mathbb{R}^n$, $f = (f^1, \ldots, f^m) : A \to \mathbb{R}^m$ is continuous at p iff each f^k is continuous at p.

(c) Any polynomial map $f : \mathbb{R}^n \to \mathbb{R}$ is continuous (built from projections via sums/products).

Exercise 2.4

Let $E \subset \mathbb{R}^n$ be compact and $f : E \to \mathbb{R}^m$ continuous.

(a) $f(E)$ is bounded. *Proof.* Each component f^j attains its max/min on E.

(b) $f(E)$ is closed, hence compact. *Proof.* Use sequential closedness via compactness of E.

(c) If E is merely closed, $f(E)$ need not be closed; e.g. $f(x) = \arctan x$ on $E = \mathbb{R}$.

Exercise 2.5 (*)

(a) If $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous and U open in $\mathbb{R}^m$, then $f^{-1}(U)$ is open.

(b) If $f^{-1}(U)$ is open for every open $U \subset \mathbb{R}^m$, then f is continuous on $\mathbb{R}^n$.

Exercise 2.6

Assume $f : \mathbb{R} \to \mathbb{R}$ differentiable with $|f'(x)| \leq M$ for all x.

(a) $|f(x) - f(y)| \leq M|x - y|$ by MVT.

(b) Hence f is uniformly continuous (Lipschitz with constant M).

Exercise 2.7

Which functions are (i) continuous, (ii) uniformly continuous on their domains?

(a) $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto e^x$: continuous, not uniformly continuous.

(b) $f : (0, 1) \to \mathbb{R}$, $x \mapsto e^x$: continuous, uniformly continuous (MVT and $e^c \leq e$).

(c) $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto \sin x$: continuous, uniformly continuous ($|\sin x - \sin y| \leq |x - y|$).

(d) $f : [0, \infty) \to \mathbb{R}$, $x \mapsto \sqrt{x}$: continuous, uniformly continuous ($|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$).

f) $f : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^n$, $x \mapsto x/\|x\|$: continuous, not uniformly continuous (fails near 0).

Exercise 2.8

Let $A \subset \mathbb{R}^n$ and $f, g : A \to \mathbb{R}$.

(a) Suppose f and g are uniformly continuous and bounded on A. Then fg is uniformly continuous and bounded on A.

Solution (boundedness). If $|f| \le M_f$ and $|g| \le M_g$ on A, then $|fg| \le M_f M_g$.

Solution (uniform continuity). Fix $\varepsilon > 0$ and set $\eta = \min\Big(1, \frac{\varepsilon}{2(M_f+M_g+1)}\Big)$. By uniform continuity of f and g, there exists $\delta > 0$ such that $\|x - y\| < \delta$ implies $|f(x) - f(y)| < \eta$ and $|g(x) - g(y)| < \eta$. Then for such x, y,

$$\begin{aligned}|f(x)g(x) - f(y)g(y)| &= |[f(x) - f(y)][g(x) - g(y)] + g(y)[f(x) - f(y)] + f(y)[g(x) - g(y)]| \\ &\le \eta^2 + M_g\,\eta + M_f\,\eta \ \le\ (M_f + M_g + 1)\eta \ \le\ \varepsilon.\end{aligned}$$

Hence fg is uniformly continuous.

(b) If f and g are uniformly continuous but not necessarily bounded, fg need not be uniformly continuous.

Counterexample. On $\mathbb{R}$, take $f(x) = x$ and $g(x) = \sin x$. Both are uniformly continuous (Lipschitz). Assume $x \sin x$ were uniformly continuous. For $\varepsilon = 1$, let $\delta > 0$ be its modulus. Set $t = \min(\delta/2, 1)$ and take $y_k = k\pi$, $x_k = k\pi + t$. Then $|x_k - y_k| = t < \delta$ but

$$|x_k \sin x_k - y_k \sin y_k| = |(k\pi + t)\sin t| \ge k\pi \sin t \xrightarrow{k\to\infty} \infty,$$

contradicting uniform continuity. Thus $x \sin x$ is not uniformly continuous.

Exercise 3.1

For $i \in \mathbb{N}$, define $f_i : [0, 1] \to \mathbb{R}$ by

$$f_i(x) = \begin{cases} 2^i x, & 0 \le x < 2^{-i}, \\ 2 - 2^i x, & 2^{-i} \le x < 2^{-(i-1)}, \\ 0, & x \ge 2^{-(i-1)}. \end{cases}$$

(a) *Sketch.* f_i is a triangular "spike" supported on $[0, 2^{-(i-1)}]$: it rises linearly from 0 at $x = 0$ to height 1 at $x = 2^{-i}$, then decreases linearly back to 0 at $x = 2^{-(i-1)}$; afterwards it is identically 0. In particular, $\sup\limits_{x\in[0,1]} f_i(x) = 1$ for every i.

(b) *Pointwise limit.* For any fixed $x > 0$ choose I such that $2^{-(i-1)} < x$ for all $i \ge I$; then $f_i(x) = 0$ for $i \ge I$. Also $f_i(0) = 0$ for all i. Hence $f_i(x) \to 0$ for every $x \in [0, 1]$.

(c) *Interchange of* lim *and* sup. We have

$$\lim_{i\to\infty} \sup_{x\in[0,1]} f_i(x) = \lim_{i\to\infty} 1 = 1, \qquad \text{while} \qquad \sup_{x\in[0,1]} \lim_{i\to\infty} f_i(x) = \sup_{x\in[0,1]} 0 = 0.$$

Therefore the equality

$$\lim_{i\to\infty} \sup_{x\in[0,1]} f_i(x) = \sup_{x\in[0,1]} \lim_{i\to\infty} f_i(x)$$

is *false* for this family. (This also shows $\{f_i\}$ does not converge uniformly to 0.)

Exercise 3.2

State whether the sequence $(f_i)_{i=0}^{\infty}$ of functions converges pointwise and/or uniformly.

(a) $f_i : \mathbb{R} \to \mathbb{R}, \quad f_i(x) = e^x + \dfrac{1}{i+1}.$

Solution. For each x, $f_i(x) \to e^x$. Moreover

$$\sup_{x\in\mathbb{R}} |f_i(x) - e^x| = \sup_{x\in\mathbb{R}} \frac{1}{i+1} = \frac{1}{i+1} \xrightarrow{i\to\infty} 0,$$

so $f_i \to e^x$ *uniformly* on $\mathbb{R}$.

(b) $f_i : \mathbb{R} \to \mathbb{R}, \quad f_i(x) = x + 2^{-i}x^2.$

Solution. For fixed x, $f_i(x) \to x$. Howeverr

$$\sup_{x\in\mathbb{R}} |f_i(x) - x| = \sup_{x\in\mathbb{R}} 2^{-i}x^2 = +\infty \quad \text{for every fixed } i,$$

hence the convergence is *not* uniform on $\mathbb{R}$. (Nonetheless, on any bounded interval $[-M, M]$ the convergence is uniform since $\sup_{|x|\le M} |f_i(x) - x| = 2^{-i}M^2 \to 0$.)

(c) $f_i : [0, 1] \to \mathbb{R}$ given by

$$f_i(x) = \begin{cases} 2^i x, & 0 \le x < 2^{-i}, \\ 2 - 2^i x, & 2^{-i} \le x < 2^{-(i-1)}, \\ 0, & x \ge 2^{-(i-1)}. \end{cases}$$

Solution. For every $x \in [0, 1]$, $f_i(x) \to 0$ (as in Exercise 3.1). But $\sup_{x\in[0,1]} f_i(x) = 1$ for all i, so the convergence is *not* uniform on $[0, 1]$. (On $[\delta, 1]$ with $\delta > 0$, the convergence is uniform.)

Exercise 3.3

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^{\infty}$ with $f_i : A \to \mathbb{R}^m$. Classify whether each statement is equivalent to (i) pointwise convergence $f_i \to f$, (ii) uniform convergence $f_i \to f$ uniformly, or (iii) neither.

(a) Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $i \ge N$:

$$\sup_{x\in A} \|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (ii) uniform convergence. This is exactly the definition via the sup norm.

(b) For all $x \in A$, for all $i \in \mathbb{N}$ there exists $\varepsilon > 0$ such that

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) neither. The statement is always true (take, e.g., $\varepsilon = \|f_i(x)-f(x)\|+1$), so it does not characterize convergence.

(c) There exists $N \in \mathbb{N}$ such that for all $x \in A$, $i \ge N$, $\varepsilon > 0$:

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) neither. Since the left-hand side is nonnegative, the inequality for *all* $\varepsilon > 0$ forces $\|f_i(x) - f(x)\| = 0$; i.e. there is N with $f_i \equiv f$ on A for all $i \ge N$. This condition is much stronger than uniform convergence and is not equivalent to it in general.

*d) $\forall \varepsilon > 0\ \forall x \in A\ \exists N \in \mathbb{N}\ \forall i \geq N :\ \|f_i(x) - f(x)\| < \varepsilon$.
Classification: (i) pointwise convergence. The quantifier order matches the definition; N may depend on x and ε.

*e) $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall x \in A\ \forall i \geq N :\ \|f_i(x) - f(x)\| < \varepsilon$.
Classification: (ii) uniform convergence. Here N works simultaneously for all $x \in A$.

*f) $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall x \in A :\ (\, i \geq N \iff \|f_i(x) - f(x)\| < \varepsilon\,)$.
Classification: (iii) neither. This is far stronger than uniform convergence (it also demands that for $i < N$ the error is $\geq \varepsilon$ for every x), and it need not hold even when $f_i \to f$ uniformly.

Exercise 3.4

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^{\infty}$ with $f_i : A \to \mathbb{R}^m$ be a sequence of functions. Assume $f_i \to f$ *uniformly* on A. Show that if $B \subset A$, then $f_i \to f$ uniformly on B.

Solution. By uniform convergence on A, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $i \geq N$ and all $x \in A$ we have $\|f_i(x) - f(x)\| < \varepsilon$. Since $B \subset A$, this inequality holds for all $x \in B$ as well, with the *same* N. Hence $f_i \to f$ uniformly on B. Equivalently,

$$\sup_{x \in B} \|f_i(x) - f(x)\| \leq \sup_{x \in A} \|f_i(x) - f(x)\| \xrightarrow{i \to \infty} 0.$$

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